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The interplay between muscle length, range of motion, and exercise selection: a review

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ABSTRACT

Emerging evidence suggests that training at longer muscle lengths may optimize muscle hypertrophy. Of note, there is an interplay between muscle length, range of motion (ROM), and exercise selection. It is difficult to identify whether muscle length per se is the main factor, limiting the ability to draw practical conclusions. Thus, synthesizing these different lines of investigation may help to clarify this topic. Firstly, we offer a comprehensive understanding of the physiological effects arising from muscle contractions in longer, moderate, and shorter muscle lengths. Subsequently, we conducted an unstructured review of long-term studies comparing muscle hypertrophy following: 1) isometric training at different muscle lengths, 2) training with different ROMs, and 3) exercises that train muscles at different muscle lengths and/or with different resistance profiles. Different lines of investigation suggest that muscle length plays a role, as training at longer muscle lengths elicits more favorable muscle growth. Notably, the greater muscle growth often observed after exercises and ROMs that train muscles at longer lengths occurs when there is a relevant external torque in the lengthened position. From a practical perspective, the selection of exercises and ROMs that impose relevant external torque at longer muscle lengths should be considered to optimize muscle growth.

Introduction

Muscle growth is an adaptation that occurs predominantly through increases in the volume of single muscle fibers [1, 2]. Mechanical tension is the main driver for inducing muscle growth. When muscle fibers detect mechanical loading, they convert the stimuli into cellular signaling, leading to an increase in muscle protein synthesis. As protein synthesis exceeds protein degradation, the phenomenon of muscle hypertrophy occurs [1, 2]. In parallel, some suggest that metabolic stress might play an indirect role in muscle growth by influencing other factors associated with hypertrophy, such as increased motor unit recruitment, metabolite production and their related physiological effects. However, this aspect is currently under debate [3–5]. Although the mechanisms of muscle hypertrophy are not yet fully elucidated, in recent years researchers have been striving to understand which factors can influence and potentially enhance muscle growth. For example, increased attention has been paid to the potential role of muscle length, range of motion (ROM), and exercise selection on hypertrophic response [6–8].

There is evidence that manipulating these factors may influence the magnitude of muscle hypertrophy [7, 9, 10], and altering muscle length has been considered the primary factor driving this differential response [11]. From a physiological standpoint, contractions performed at shorter muscle-tendon unit lengths—which are commonly associated with more flexed joint positions—may involve suboptimal sarcomere overlap, potentially reducing cross-bridge formation and force output [12]. However, it is important to note that joint-level torque is also modulated by other factors, including muscle moment arms, tendon compliance, activation strategies, and the load–velocity profile, which collectively influence the mechanical output during movement. In contrast, contractions occurring at moderate muscle lengths (ML) are generally associated with more favorable sarcomere overlap, potentially allowing for a greater number of effective cross-bridge interactions and improved active force-generating capacity [13, 14]. Similarly, contractions performed at longer muscle lengths (LL) may, in certain contexts—particularly during active lengthening—be associated with increased overall force production, which can be partially attributed to the contribution of passive elastic components such as titin [15]. Thus, in theory, contractions performed at different joint angles may expose muscle fibers to varying levels of mechanical tension, potentially leading to distinct hypertrophic stimuli.

One way to investigate the effects of contractions performed at different muscle lengths on hypertrophy is through isometric training at varying joint angles. This approach allows researchers to isolate the influence of muscle length more directly, minimizing confounding factors related to joint movement or varying torque-angle curves. Previous studies suggest that isometric training at LL leads to more favorable hypertrophy compared to SL [16, 17], though some studies found no significant differences [18, 19]. Thus, a question arises: could there be a muscle length "threshold" at which muscles start to exhibit appreciable differences in growth, as mentioned earlier [20]? The analysis of studies that compared other muscle lengths, such as at ML, would help to reinforce or refute this reasoning. The latest review on the topic is not recent and is limited to a few studies that compare isometric training at SL and LL, making it difficult to understand the influence of muscle length on the hypertrophic response [9]. Furthermore, the relationship between muscle length and hypertrophy gains more complexity when dealing with dynamic contractions involving variations in the ROM, necessitating deeper analysis.

The effect of different ROM configurations on muscle hypertrophy is currently a hot topic [6, 10, 21]. Historically, full ROM has been considered the standard for optimizing muscular adaptations [22], with the literature suggesting, until recently, that full ROM is superior to partial ROM for inducing muscle hypertrophy [6]. However, emergent studies comparing training with different partial ROMs (e. g., initial vs. final portion of ROM) have allowed exercise scientists to revisit this notion [23, 24]. For example, it has been observed that both full ROM and initial ROM training (i. e., when muscle contracts exclusively at LL) are superior to training at the final ROM (i. e., when muscle contracts exclusively at SL) [25, 26]. In addition, more favorable hypertrophy has been observed after training at the initial ROM [24, 27, 28], or at the intermediary ROM (i. e., when muscle contracts near the ML: inter ROM) compared to full ROM [29]. These observations are apparently counterintuitive because initial ROM and inter ROM are a part contained in full ROM. Therefore, ROM alone does seem not to explain the differences between full ROM and another ROM configuration. It could be that other factors are modulating hypertrophic adaptation, such as external torque, muscle length, or the interaction between them.

Investigations comparing different resistance exercises offer a unique opportunity to identify the potential role of external torque, muscle length, and interaction between them. It is possible to manipulate the relative length at which a muscle is overloaded during training by selecting exercises that alter joint angles (e. g., shoulder in neutral vs. hyperextended position) and/or by placing the peak torque at different portions of the ROM—for example, overloading the muscle in the initial phase of the concentric action, when it is at a more elongated position, versus the final phase, when it is more shortened [30]. While such variations may not always place the muscle on the descending limb of the length-tension curve—for example, the biceps brachii [31]—they can still represent meaningful differences in the mechanical environment experienced by the fibers. There are reports that resistance exercises emphasizing the initial, more elongated portion of the ROM may optimize hypertrophic adaptations compared to those emphasizing SL [32]. However, others fail to observe this superiority [8]. For example, performing the biceps curl at LL without relevant external torque in this position does not seem to induce greater adaptations compared to the biceps curl at SL [8]. Thus, training with LL exercise per se may not be sufficient for optimizing muscle growth; there should probably also be concomitant external torque at the lengthened position.

Given these nuances, a review that synthesizes and compiles these different lines of investigation (i. e., muscle length, ROM, and exercise selection literature) would help to clarify this topic, draw practical conclusions, and highlight current research gaps. Thus, firstly, we aimed to present a physiological background related to contractions at different muscle lengths and then perform a synthesis of training studies that compare the hypertrophic effects of 1) isometric training at different joint angles, 2) training with different ranges of motion (ROMs), and 3) resistance exercises that impose peak torque in differ-

ent portions of ROM and/or train muscles at different joint angles/muscle length.

Experimental approach to the problem

The methodology of the present study is divided into two parts. The first part is a comprehensive analysis of the physiological effects resulting from contractions performed at different joint angles. Subsequently, three unstructured reviews were undertaken regarding hypertrophic response following training. The first focuses on isometric training at different joint angles, the second encompasses training with different ROMs, and the third explores different resistance exercises that impose peak torque at varying portions of ROM. When searching for exercise selection studies, we chose to include exclusively studies comparing head-to-head single-joint resistance exercises. This decision-making is because multi-joint resistance exercises differ from single-joint exercises in many other factors besides muscle length and external torque (e. g., number of muscles involved, stability requirement, complexity, two-joint muscle principle) which could bias the establishment of conclusions. For a comprehensive review of the role of exercise selection, see previous references [7, 33].

The unstructured reviews were conducted by the three authors (GP, MM, and WK) using the PubMed and Scopus databases as well as seminal textbooks to identify studies that investigated the three topics cited above (isometric training in different joint angles and hypertrophy; training with different ROMs and hypertrophy; training with different resistance exercises and hypertrophy). The reviews sought references to terms specific to the topics, such as isometric training, joint angle, resistance exercise, range of motion, exercise selection, torque, hypertrophy, muscle volume, cross-sectional area, and muscle thickness.

Physiological background

Changes in joint angles influence muscle-tendon unit lengths, however, this relationship is not straightforward and becomes increasingly complex when involving multi-joint resistance exercises and muscles that span two joints [13, 34]. Within this context, concentric muscle contractions progressively shorten the muscle fiber with each attained angle. Theoretically, muscle fiber shortening likely promotes an increase in the radial distance between the actin and myosin protein filaments (lattice space), which may decrease the interaction of calcium (Ca^{2+}) with troponin, thereby diminishing the muscle's capacity to generate active force [12].

On the other hand, if muscle contraction occurs within a determined ROM where the muscle fiber is more stretched, there is a reduction in the overlap between actin and myosin filaments, likely resulting in diminished active muscle force production capacity [12]. This relationship between muscle fiber length and active force production was initially described by Gordon et al. [35] and is represented by the length-tension relationship. Notably, during a contraction with a maximal voluntary effort at LL, the transmission of force along the muscle-tendon unit (from origin to insertion) might be facilitated by an increase in its stiffness, in which the force produced by contractile elements overlaps the tension generated by passive elements, particularly in the presence of Ca^{2+} [36]. Thus, it is speculated that the maximum tension produced by the muscle fiber is higher when the muscle contracts at LL compared to when it contracts at SL [12].

Furthermore, when the muscle fiber is near its resting length (i. e., ML), limitations in force production and transmission by the muscle-tendon unit are not as pronounced. This is likely due to the optimal overlap of myosin thick filaments and actin thin filaments, enabling the formation of the maximum possible number of force-generating cross-bridges [12]. Therefore, it is acceptable to suppose that this condition represents the optimal length for active force production [35].

This difference in the muscle fiber's ability to generate tension when contracted at LL, SL, and ML may influence the magnitude of mechanical tension and sensitize the mechanosensor present in the muscle cell, consequently affecting hypertrophic stimulus [37]. This line of evidence is supported by Russ [38], who utilized rats (in situ) to compare force production and Akt activation after contractions via electrical stimulation at LL, ML, and SL. The results showed that force production and Akt activation were higher when the muscle contracted at ML, followed by LL rather than SL, in which values were similar to the sham condition (no stimulation) [38]. It is interesting to emphasize the increase in passive tension resulting from stretching contributed to greater Akt activation compared to the condition where the muscle shortened when electrically stimulated in the sham condition [38]. It is worth mentioning that there are indications that titin (partly responsible for passive force production) is one of the mechanosensors linked to triggering cascades of events directed toward muscle hypertrophy [39]. This means that mechanical tension from passive elements, occurring in lengthened muscle positions, could contribute to enhancing muscle hypertrophy stimulus.

In agreement, Van Dyke et al. [40] observed that the phosphorylation levels of mechano-responsive kinases (Akt and p70S6K) seem to be contingent upon the muscle's length, with contractions at LL and ML enhancing this response. The results of Russ [38] and Van Dyke et al. [40] suggest that the physiological responses linked to muscular hypertrophy occur to a greater extent at ML or LL when compared to those occurring at SL. From a practical standpoint associated with ROM manipulation and isometric training, contractions performed in the final ROM, where the muscle contracts at an SL, may be not sufficient to activate hypertrophic pathways to the same extent as contractions performed in the full, initial, and inter ROMs, where the muscle contracts at LL and ML. Furthermore, in addition to mechanical tension, the manipulation of the ROM also appears to modulate metabolic stress, which has been suggested as a factor that may influence muscle hypertrophy [3].

Mizuno et al. [41] observed that the oxygenation level in the vastus lateralis muscle during isometric contractions in the knee extension exercise at 90° knee flexion (0° = knee extended) (i. e., at LL) was lower than the level found during contractions at 30° knee flexion (i. e., at SL). These findings were corroborated by other studies, which have shown that oxygen consumption was lower when contractions occurred at 30° knee flexion compared to 60° and 90° knee flexion [37, 42]. Furthermore, Kooistra et al. [43] observed that the time taken to reach maximal oxygen consumption was longer in isometric contractions at 30° knee flexion than at 60° and

▶ **Table 1** Summary of the studies on the effects of isometric training at different muscles lengths on muscle hypertrophy.

Study	Sample	Training protocol	Joint angle	Muscles assessed	Overall hypertrophy effect
Alegre et al. [17]	Untrained young adults (n = 19)	8 wks, 2–3x/wk, 3–4 sets of 5–7 reps of 5 s at 60–80 % of MVC	Knee flexion: SL: 50°, LL: 90°	Vastus lateralis muscle thickness at proximal, middle, and distal sites	LL induced greater increases at middle (LL: 13.5 %, SL: 5.2 %), but similar at proximal (LL: 9.3 %, SL: 6.1 %) and distal (LL: 9.0 %, SL: 9.0 %) sites
Akagi et al. [19]	Untrained young adults (n = 26)†	8 wks, 3x/wk, 3 sets of 8–10 reps of 5-s MVC	Plantar flexion: SL: 0°, LL: 40°	Tibialis anterior muscle thickness	Both groups increased tibialis anterior (LL: 1.5 %, SL: 0.9 %) with no between-group differences
Kubo et al. [18]	Untrained young men (n = 18)†	12 wks, 4x/wk, 6 sets of 15 s at 50–70 % of MVC	Knee flexion: SL: 50°, LL: 100°	Volume of whole and individual quadriceps heads	Similar increases of whole quadriceps (LL: 11.0 %, SL: 10.0 %), vastus lateralis (LL: 11.7 %, SL: 10.5 %), vastus medialis (LL: 13.3 %, SL: 13.5 %), and rectus femoris (LL: 9.1 %, SL: 9.0 %), and vastus intermedius (LL: 12.0 %, SL: 10.5 %)
Nakao et at. [51]	Untrained young adults (n = 28)	8 wks, 4x/wk, 5 sets of 20 reps of 5 s at 30 % of MVC	Knee flexion: SL: 90°, LL: 30°	Hamstrings CSA at proximal, middle, and distal sites	Similar increases of hamstrings at proximal (LL: 3.9 %, SL: 6.2 %), middle (LL: 6.2 %, SL: 5.3 %), and distal (LL: 5.4 %, SL: 5.2 %) sites
Noorkoiv et al. [16]	Untrained young men (n = 16)	6 wks, 3x/wk, 5 sets of 5 s MVC	Knee flexion: SL: ~38°, LL: ~87°	Volume and CSA of whole and individual quadriceps heads	Only LL experienced increases of whole quadriceps (LL: 5.2 %, SL: –0.5 %), vastus lateralis (LL: 6.3 %, SL: –1.2 %), vastus medialis (LL: 4.8 %, SL: –1.2 %), and rectus femoris (LL: 8.2 %, SL: 0.0 %), but not vastus intermedius (LL: 3.1 %, SL: 0.5 %)

Abbreviations: MVC = maximum voluntary contraction, SL = shorter muscle length, LL = longer muscle length, CSA = cross-sectional area. † = within-subject design.

90°. Taken together, these findings indicate a reduced metabolic demand when the muscle contracts at joint angles associated with an SL in comparison to ML and LL. It is worth noting that the accumulation of metabolites in the muscle is thought to trigger events associated with the development of muscular hypertrophy [3, 44], although we are unaware of empirical evidence that this may be the case. In fact, such accumulation may primarily accelerate the onset of fatigue without directly enhancing hypertrophic signaling—a hypothesis that warrants further investigation to be clarified.

Additionally, the transitory change in muscle size (often referred to as muscle swelling), which may occur shortly after or during resistance exercise, has also been interpreted as a factor associated with metabolic stress [45, 46]. In the study by Fouré et al. [47], muscle swelling, measured by the increase in volume of the quadriceps muscles, was compared between the right and left lower limbs following an electrostimulation session. In each lower limb, the vastus lateralis and vastus medialis muscles underwent the same electrostimulation protocol at two different joint angles: LL and SL (100° and 30° knee flexion, respectively, with 0° representing a full knee extension). The study revealed that muscle swelling was more pronounced when the muscles were stimulated at the LL.

Interestingly, Hirono et al. [48] found a positive and significant correlation between the magnitude of quadriceps femoris muscle swelling after the first training session and the magnitude of training-induced muscle growth. In this sense, it is possible to hypothesize that metabolic stress may play an indirect role in hypertrophy and if this is the case, LL appears to be related to greater metabolic stress. Some argue that the presence of metabolites would contribute to increased recruitment of motor units [44], particularly when training at LL compared to SL [49]. Since the recruited motor units are the ones prone to adapting through hypertrophy [50], the increased recruitment from contractions at longer muscle lengths (LL) compared to shorter muscle lengths (SL) may be the mediating factor of the relationship between metabolic stress indicators and signaling of factors related to muscle hypertrophy. While this explanation is plausible, further research is needed to verify it.

Isometric training at different muscle lengths

Five studies comparing muscle hypertrophy after isometric training at different joint angles, and consequently muscle lengths, were identified. The summarized information from these studies is presented in ▶ **Table 1**.

Alegre et al. [17] compared isometric knee extension training at SL and LL and observed more favorable quadriceps hypertrophy for the LL group, particularly in the central portion of the vastus lateralis. In agreement, Noorkõiv et al. [16] compared isometric knee extension training at SL and LL, and only LL experienced hypertrophy of the whole quadriceps and in three of the four individual muscle heads. On the other hand, Kubo et al. [18] observed hypertrophy of the whole quadriceps and individual heads, but it was not different between isometric knee extension training performed at SL and LL. Perhaps such dissimilarities may be due to the muscle length investigated. Among the three studies, the study that used the shorter muscle length was that of Noorkõiv et al. [16] (~38 ° of knee flexion; 0 ° full knee extension) and was the one that showed the smallest relative changes in the SL group (–1.2 to 0.5 %). The other groups from other studies trained at ≥ 50 ° of knee flexion, suggesting that there may be a threshold of muscle length necessary to elicit appreciable quadriceps hypertrophy.

Nakao et al. [51] investigated isometric knee flexion training at SL and LL, with the hip joint angle maintained at 90 ° in both groups. Hamstring hypertrophy of the proximal, middle, and distal sites did

not differ between groups. Of note, the hamstrings are biarticular muscles (except the biceps femoris short head) that cross the hip and knee joints. Given that the hip was flexed at 90°, it is likely that in both groups the hamstrings were trained at ML to LL, which may have resulted in optimal stimulation in both groups, contributing to similar hypertrophy. Akagi et al. [19] investigated the effect of isometric dorsiflexion training at LL and SL on tibialis anterior muscle size and did not observe different changes between the groups. Of note, the changes observed in both groups were minimal (0.9% and 1.5%, SL and LL, respectively). This hampers the ability to identify potential differences between muscle lengths.

Overall, based on these isometric training findings, it is possible to infer that muscle length plays a role in muscle growth and that perhaps there is a muscle length "threshold" from which muscles start to experience appreciable growth. It is emphasized that further investigations are needed in other muscles and at varying joint angles and muscle lengths, particularly using ML. If there is, in fact, a muscle length threshold that enhances hypertrophy, because it is a joint angle, for example, of the higher internal moment arm of a given muscle, this condition might enable even greater gains by reaching the muscle length threshold, potentially amplifying physiological responses due to a greater internal muscle force capacity [37, 38, 52]. Future investigations should explore this possibility.

ROM comparisons

We found a total of 13 studies comparing muscle size changes after training with different ROM configurations. The summarized information from these studies is presented in ▶ **Table 2**. Six studies included the comparison of full ROM vs. final ROM, two studies included a comparison between full ROM vs. inter ROM, five studies included a comparison between initial ROM vs. final ROM, and five included a comparison between initial ROM vs. full ROM.

Full ROM vs. final ROM

Bloomquist et al. [25] compared the hypertrophic effects of squats performed with full vs. final ROM and observed greater quadriceps growth in five of the six sites for full ROM training. In agreement, McMahon et al. [53] compared several lower limb resistance exercises performed with full vs. final ROM and observed greater vastus lateralis muscle growth at the proximal and distal sites, but not the middle, after full ROM training. Pedrosa et al. [27] compared the effects of performing knee extension with full vs. final ROM in the vastus lateralis and rectus femoris muscle size changes. The authors observed greater muscle growth in the most distal sites after full ROM training, with no differences at the most proximal sites. Collectively, these findings suggest that achieving longer muscle lengths through full ROM training optimizes muscle growth, especially at more distal muscle sites. Of note, this observation is not universal [54, 55]. Therefore, this hypothesis needs additional studies. Partially in agreement with observations that full ROM induces more favorable hypertrophic responses, Kassiano et al. [24] compared the effect of calf raises performed with full ROM vs. final ROM on gastrocnemius hypertrophy. The authors found that only full ROM training induced growth in both gastrocnemius heads, while final ROM training only increased the size of the lateral gastrocnemius [24].

Notably, others fail to observe superior gains after full ROM training [54, 55]. Kubo et al. [54] compared the effect of squats with full vs. final ROM and observed similar hypertrophy of the three vastus muscles (the rectus femoris did not show appreciable muscle size changes, regardless of the group). Interestingly, the group considered as the final ROM in the Kubo et al. [54] study performed squats from 0° to 90° of knee flexion, similar to the full ROM group (0° to 90°; see ▶ **Table 2**) in the McMahon et al. [53] study. Together, these observations reinforce the previously mentioned notion that there is likely a muscle length threshold from which the muscle experiences significant hypertrophy and that exceeding this threshold may not provide additional benefits. Notably, the full ROM showed greater hypertrophy in the gluteal muscles and hip adductors compared to the final ROM group. Perhaps, these muscles did not reach the length threshold that enhances hypertrophy during training with the final ROM. Although reasonable, this hypothesis requires scrutiny, among other things, because the response of greater hypertrophy may depend on exercise selection.

Additionally, Valamatos et al. [55] found no significant difference in vastus lateralis hypertrophy between groups training with full vs. final ROM using an isokinetic dynamometer, potentially because the final ROM group performed more sets to equate time under tension with the full ROM group, thereby minimizing potential disparities in hypertrophic outcomes. Thus, it is not possible to know what the results would be like if the number of sets had been similar between the groups. Also, such non-superiority of full ROM over final ROM may be related to the more constant external torque that the isokinetic dynamometer imposes throughout the ROM [56]. The same is not necessarily true for conventional knee extension machines. Some knee extension machines actually decrease or increase the external torque in the final portion of the movement due to a manufactured cam system [57]. If accepted, this hypothesis implies that muscle length may not be the only factor in optimizing muscle growth. Perhaps it is an interaction between external torque and muscle length. This hypothesis will be expanded thereafter, mainly in the exercise comparison section.

Full ROM vs. inter ROM

Pinto et al. [58] compared biceps preacher curl performed with full ROM vs. inter ROM and observed a similar increase in elbow flexor muscle size between the groups. When analyzing the schematic figure presented by the authors, it is possible to notice that both groups trained in the portion of the ROM that is the peak of torque; ~45° to 50° of elbow flexion where likely occurred the greater external moment arm (right angle between resistance [i. e., barbell] and the line of gravity). Such aspects may have contributed to the similarity between groups, even though the inter ROM did not reach the very initial portion of ROM (i. e., 0° to 49°). Goto et al. [29] compared lying elbow extension with full ROM vs. inter ROM and observed greater hypertrophy of the triceps brachii after inter ROM. Once again, the portion of ROM where peak torque occurred was present in both groups; i. e., ~90° of elbow flexion, where the greater external moment arm likely occurred. This would explain similar gains but not greater muscle hypertrophy in the inter ROM group. Some could argue a higher external torque in the inter ROM group due to lifting heavier weights, as observed in partial ROM groups in other studies

▶ **Table 2** Summary of studies comparing the hypertrophic effects of different ROM configurations.

Study	Sample	Training protocol	Exercise/ROM	Muscles assessed	Muscle hypertrophy effect
Full ROM vs. final ROM					
Bloomquist et al. [25]	Untrained young men	12 wks, 3x/wk, 3–5 sets of 3–10RM	Squat: final ROM: 0°–60°, full ROM: 0°–120°	CSA of the anterior and posterior thigh at six proximo-distal sites and vastus lateralis muscle thickness at middle site	Full ROM induced greater anterior thigh increases at five of six sites (full ROM: 4.3–6.1%, final ROM: −2.3–2.3%). The most proximal site did not differ (full ROM: 7.1%, final ROM 4.6%) Changes in posterior thigh did not differ between groups (−1.2–3.0% vs. −1.3–2.7%) Changes in vastus lateralis muscle thickness did not differ between groups (full ROM: 3.6%, final ROM: 2.8%)
Kassiano et al. [24]	Untrained young women	8 wks, 3x/wk, 3 sets of 15–20RM	Calf raise horizontal leg press, final ROM: 0°– + 25°, full ROM: −25°– + 25°	Muscle thickness of gastrocnemius medial and lateral	Both groups increased the lateral gastrocnemius (full ROM: 7.3%, final ROM: 6.2%), but only full ROM increased the medial gastrocnemius (full ROM: 6.7%, final ROM: 3.4%)
Kubo et al. [54]	Untrained young men	10 wk, 2x/wk, 3 sets at 60–90% 1RM	Squat: final ROM: 0°–90°, full ROM: 0°–140°	Muscle volume of rectus femoris, vastus lateralis, vastus intermedius, vastus medialis, biceps femoris short head, biceps femoris long head, semitendinosus, semimembranosus, adductor magnus, adductor longus, adductor brevis and gluteus maximum	Full ROM induced greater increases in size of gluteus maximus (full ROM: 6.4%, final ROM: 2.1%) and adductors (full ROM: 5.1%, final ROM: 2.7%), while the increases in vastii were similar (full ROM: 3.5–6.8%, final ROM: 4.8–6.7%); rectus femoris, biceps femoris short and long heads, semitendinosus and semimembranosus did not change in either group (full ROM: −0.3–1.3%, final ROM: −0.5–1.0%)
McMahon et al. [53]	Untrained young adults	8 wks, 3x/wk, 3 sets of 8–10 reps at 80% 1RM	Squat, knee extension, Bulgarian split squat, double- and single-legged Samson chair, leg press, and lunge: final ROM: 0°–50° full ROM: 0°–90°	CSA of vastus lateralis at proximal, middle, and distal sites	Full ROM induced greater CSA increases at proximal (full ROM: 33.8%, final ROM: 19.0%) and distal (full ROM: 40.1%, final ROM: 7.5%) sites; similar increases were observed at middle site (full ROM: 18.0%, final ROM: 22.0%)
Pedrosa et al. [27]	Untrained young women	12 wks, 3x/wk, 3–6 sets of 7 reps at 60% 1RM	Knee extension: final ROM: 65°–30°, full ROM: 100°–30°	CSA of vastus lateralis and rectus femoris at four proximo-distal sites	Full ROM induced greater increases in the rectus femoris at 60% (full ROM: 18.5%, final ROM: 9.1%) and 70% (full ROM: 17.7%, final ROM: 1.2%) of the thigh length. Similarly, full ROM induced greater hypertrophy in the vastus lateralis at 60% (full ROM: 10.7%, final ROM: 4.8%) and 70% (full ROM: 10.2%, final ROM: 1.1%) of the thigh length. There were no differences between groups at the other sites (full ROM: 9.7–15.4%, final ROM: 9.1–14.6%)
Valamatos et al. [55]	Untrained men	15 wks, 3x/wk, 2–7 sets, 6–15RM	Knee extension (isokinetic): final ROM: 0°–60°, full ROM: 0°–100°	Volume and CSA of vastus lateralis at proximal, middle, and distal sites	Similar increases in volume (full ROM: 7.6%, final ROM: 6.7%) and CSA at proximal (full ROM: 3.0%, final ROM: 3.0%), middle (full ROM: 5.4%, final ROM: 4.5%), and distal (full ROM: 6.9%, final ROM: 6.6%) sites

▶ **Table 2** Continued.

Study	Sample	Training protocol	Exercise/ROM	Muscles assessed	Muscle hypertrophy effect
Full ROM vs. inter ROM					
Goto et al. [29]	Resistance trained men	8 wks, 3x/wk, 3 sets of 8RM	Barbell lying elbow extension: full ROM: 0°–120°, inter ROM: 45°–90°	CSA of the triceps brachii	Inter ROM induced greater increases in CSA of the triceps brachii (inter ROM: 49.5 %, full ROM: 27.6 %)
Pinto et al. [58]	Untrained men	10 wks, 3x/wk, 2–4 sets, 8–20RM	Barbell preacher curl: full ROM: 0°–130°, inter ROM: 50°–100°	Muscle thickness of elbow flexors	Similar muscle thickness increases (full ROM: 7.2 %, inter ROM: 7.8 %)
Full ROM vs. initial ROM					
Kassiano et al. [24]	Untrained women	8 wks, 3x/wk, 3 sets of 15–20RM	Calf raise (horizontal leg press: initial ROM: –25–0°, full ROM: –25°– + 25°	Muscle thickness of gastrocnemius medial and lateral	Initial ROM induced greater increases in medial gastrocnemius (initial ROM: 15.2 %, full ROM: 6.7 %), but similar increases in lateral gastrocnemius (initial ROM: 14.9 %, full ROM: 7.3 %)
Pedrosa et al. [27]	Untrained women	12 wks, 3x/wk, 3–6 sets of 7 reps at 60 % 1RM	Knee extension: initial ROM: 100°–65°, full ROM: 100°–30°	CSA of vastus lateralis and rectus femoris at four proximo-distal sites	Initial ROM induced greater increases in the rectus femoris at 40 % (initial ROM: 20.9 %, full ROM: 9.7 %), 50 % (initial ROM: 24.9 %, full ROM: 13.0 %), and 70 % (initial ROM: 30.2 %, full ROM: 17.7 %), but not at 60 % (initial ROM: 18.5 %, full ROM: 21.6 %) Initial ROM induced greater increases in the vastus lateralis at 70 % (initial ROM: 15.6 %, full ROM: 10.2 %), but not at 40 % (initial ROM: 12.2 %, full ROM: 15.4 %), 50 % (initial ROM: 16.0 %, full ROM: 12.4 %), and 60 % (initial ROM: 14.7 %, full ROM: 10.7 %)
Werkhausen et al. [60]	Resistance-trained young adults	10 wks, 3x/wk, 4–6 sets of 4–6 RMs	Leg press: initial ROM: 81°–90°, full ROM: 0°–90°	Muscle thickness of vastus lateralis	No changes in either group (initial ROM: –1.2 %, full ROM: –2.1 %)
Wolf et al. [63]	Trained men	8 wks, 2x/wk, 4 sets of 5–10RM	Flat machine chest press, bench dumbbell row, dumbbell overhead triceps extensions, dumbbell supinating curl, incline machine chest press, cable single arm pulldown, cable pushdown, Bayesian curl: initial ROM: ~50 % of full ROM; full ROM: all ROM individual available.	Muscle thickness of elbow flexor and extensor at 45 % and 55 % of humeral length	Initial ROM and full ROM resulted in similar hypertrophy across all muscles and regions. For the elbow flexors at 55 %, initial ROM led to ~5 % growth, and full ROM also resulted in ~5 %. At 45 %, initial ROM produced ~4 % growth, while full ROM led to ~3 %. For the elbow extensors at 55 %, initial ROM resulted in ~5 % growth, while full ROM showed a more pronounced hypertrophy of ~8.7 %. At 45 %, initial ROM achieved ~6.2 % growth, with full ROM slightly higher at ~7.5 %.
Initial ROM vs. final ROM					

▶ **Table 2** Continued.

Study	Sample	Training protocol	Exercise/ROM	Muscles assessed	Muscle hypertrophy effect
Kassiano et al. [24]	Untrained women	8 wks, 3x/wk, 3 sets of 15–20RM	Calf raise horizontal leg press: initial ROM: –25–0°, final ROM: 0°– + 25°	Muscle thickness of gastrocnemius medial and lateral	Initial ROM induced greater increases in lateral (initial ROM: 14.9 %, final ROM: 6.2 %) and medial (initial ROM: 15.2 %, final ROM: 3.4 %) gastrocnemius
McMahon et al. [26]	Untrained young adults	8 wks, 3x/wk, 3–4 sets of 8–10 reps at 55 % 1RM for initial ROM and at 80 % 1RM for final ROM	Bilateral barbell squat, leg press, knee extension, Bulgarian split squat, and Samson chair: initial ROM: 40°–90°, final ROM: 0°–50°	CSA of vastus lateralis at proximal, middle, and distal sites	Initial ROM induced greater increases at the proximal (initial ROM: 30.5 %, final ROM: 17.4 %), middle (initial ROM: 35.8 %, final ROM: 21.6 %), and distal (initial ROM: 50.7 %, final ROM: 13.5 %) sites
Pedrosa et al. [59]	Untrained women	8 wks, 3x/wk, 4 sets at 60–100 % of 1RM	Seated preacher curl: initial ROM: 0°–68°, final ROM: 68°–135°	CSA of biceps brachii at proximal and distal sites	Initial ROM induced greater increases at the distal (initial ROM: 24.0 %, final ROM: 13.1 %) but similar at the proximal (initial ROM: 20.0 %, final ROM: 21.1 %) site
Pedrosa et al. [27]	Untrained women	12 wks, 3x/wk, 3–6 sets of 7 reps at 60 % 1RM	Knee extension: initial ROM: 100°–65°, final ROM: 65°–30°	CSA of vastus lateralis and rectus femoris at four proximo-distal sites	Initial ROM induced greater increases in the rectus femoris at 40 % (initial ROM: 20.9 %, final ROM: 9.7 %), 50 % (initial ROM: 24.9 %, final ROM: 10.4 %), 60 % (initial ROM: 21.6 %, final ROM: 9.1 %), and 70 % (initial ROM: 30.2 %, final ROM: 1.2 %). Initial ROM induced greater increases in the vastus lateralis at 50 % (initial ROM: 16.0 %, final ROM: 9.1 %), 60 % (initial ROM: 14.7 %, final ROM: 4.8 %), and 70 % (initial ROM: 15.6 %, final ROM: 1.1 %), but without difference at 40 % (initial ROM: 15.4 %, final ROM: 14.6 %)
Sato et al. [64]	Untrained young adults	5 wks, 2x/wk, 3 sets of 10 reps at 30–100 % of the MVC	Preacher curl: initial ROM: 0°–50°, final ROM: 80°–130°	Muscle thickness of elbow flexors at proximal, middle, and distal sites	Initial ROM induced greater increases at distal site (initial ROM: 12.7 %, final ROM: 2.1 %), but similar at proximal (initial ROM: 5.0 %, final ROM: 3.7 %) and middle (initial ROM: 6.9 %, final ROM: 5.0 %) sites

Abbreviations: ROM = range of motion, RM = repetition maximum, CSA = cross-sectional area, MVC = maximum voluntary contraction.

[24, 27, 59], but the inter and full ROM groups in the Goto et al. [29] study did not train with different weights. Thus, it is difficult to reconcile these findings. Perhaps other factors may explain such differences, such as the lower variation in torque along the ROM, which could enhance blood flow restriction and optimize the effects of metabolic stress, as suggested by the authors [29].

Full ROM vs. initial ROM

If one accepts that the main factor modulating the magnitude of muscle hypertrophy is training at longer lengths, the most parsimonious hypothesis would be that initial ROM results in gains similar to full ROM, as both ROMs allow reaching longer muscle lengths. However, the available literature, at least in part, does not seem to support this notion. For example, Pedrosa et al. [27] compared knee extension exercise performed with full vs. initial ROM and found that training with the initial ROM resulted in greater rectus femoris and vastus lateralis hypertrophy. Similarly, Kassiano et al. [24] compared calf raises performed with full vs. initial ROM and observed greater hypertrophy of the medial gastrocnemius but without between-group differences for the lateral gastrocnemius. Maeo et al. [28] (not included in the review as it has not been published yet) compared performing a hip extension exercise with full vs. initial ROM and reported greater increases in gluteus maximus size after initial ROM training. Based on these observations, it appears that the ROM—and consequently the muscle length—is not the only factor responsible for these results. When scrutinizing such studies, it is possible to note that the initial ROM group in the study by Pedrosa et al. [27] and Kassiano et al. [24] trained with

heavier weight than the full ROM group. Therefore, the initial ROM group likely trained with a higher external torque than the full ROM group. Based on this observation, it is reasonable to propose that the combination of muscle length and greater external torque may have contributed to the more favorable results for the initial ROM training.

However, such more favorable findings in response to initial ROM are not unanimous. Werkhausen et al. [60] compared performing a leg press with full vs. initial ROM and found no significant difference between the groups in inducing vastus lateralis muscle growth. It should be noted the training groups in Werkhausen et al. [60] performed the exercise concentric-only, with three-second pauses between repetitions, and participants were instructed to perform repetitions as fast as possible. There is support that eccentric muscle actions play a role in hypertrophy [61], and pauses between repetitions may mitigate local fatigue, which is also considered an important factor in facilitating muscle hypertrophy [62]. Thus, it is possible that such training schemes have affected the magnitude of hypertrophy and hampered the ability to test whether ROM plays a role in the magnitude of muscle size changes. Indeed, no appreciable changes in the vastus lateralis size were observed in both groups of Werkhausen et al. [60] (ranging from –1.2 % to –2.1 %).

More recently, Wolf et al. [63] showed a similar hypertrophic response in the elbow flexors and extensors after 8 weeks of training with full and initial ROM using four exercises targeted for the elbow flexor and four for the elbow extensors. The training was configured to reach task failure (5–10 or 10–15 maximum repetitions), and both protocols were performed with 2–second eccentric, 1 second after eccentric, and explosive in concentric. Note that no mechanical stop or visual mark was used to control the ROM. The participants self-regulated the ROM, and all sessions were monitored by one or two supervisors. The lack of control over the ROM may have allowed participants to perform repetitions outside the desired ROM, which could impact the muscle length-tension curve as well as the physiological responses associated with hypertrophy stimulus [38, 41, 43]. Still, initial and full ROM presented similar results in hypertrophy, reinforcing the notion that hypertrophy is potentially influenced by training that provides tension at longer muscle lengths. Overall, from the results of full vs. initial ROM, it is possible to suggest that length alone is likely not the major factor. Collectively, these observations suggest that the greater muscle hypertrophy often observed in these experiments has been driven by the interaction between training at longer muscle lengths and higher external torque in that position.

Initial ROM vs. final ROM

A head-to-head comparison of initial vs. final ROM offers an opportunity to contrast training at different muscle lengths with less disparity in weight lifted (because both partial ROMs usually allow lift of heavier weight compared to full ROM) and consequently less disparity in external torque. Pedrosa et al. [27] compared performing knee extensions at initial vs. final ROM and observed greater rectus femoris and vastus lateralis hypertrophy across almost all sites analyzed after initial ROM training (the exception was the most proximal vastus lateralis site). In agreement, Kassiano et al. [24] compared performing the calf raise exercise at initial vs. final ROM and found greater hypertrophy of both gastrocnemius heads after initial ROM training. It is worth noting that in both studies [24, 27], the weight lifted by the initial and final ROM groups did not differ between them. Moreover, a greater external moment arm is expected to occur at the final compared to the initial angles of knee extension (58). Thus, in the Pedrosa et al. [27] study, initial and final ROM groups trained the leg extension with similar relative and absolute weights, which means the final ROM group trained with greater external torque, yet this condition still resulted in lower hypertrophy responses. Such findings suggest that the higher external torque, produced by lifting heavier weights, is not sufficient to affect the hypertrophy magnitude, but rather the combination of high external torque applied at longer muscle lengths, a scenario offered by the initial ROM condition.

In partial agreement, McMahon et al. [26] compared performing lower limb exercises at initial vs. final ROM and observed greater distal hypertrophy of the vastus lateralis in favor of the initial ROM, with no differences in the proximal and middle sites. Pedrosa et al. [59] compared performing the preacher curl at the initial ROM vs. final ROM and observed greater hypertrophy of the biceps brachii at the distal site but without difference at the proximal site. Sato et al. [64] also compared performing biceps preacher curl at initial vs. final ROM and observed greater hypertrophy in the distal site of the elbow flexors after initial ROM training. Similar to full vs. final ROM findings, there appears to be a certain consistency in greater distal muscle hypertrophy after training at longer muscle lengths (i. e., full and initial ROM). Also, the results of the initial ROM vs. final ROM—together with previous comparisons of different ROMs and isometric training literature—allow us to infer that the ROM variable is not the factor per se but rather a means of training the muscle at lengthened positions with relevant torque in that position.

Additionally, analyzing the exercise selection literature can serve as proof of this concept because the head-to-head comparison of different exercises allows keeping the ROM variable constant while the independent variables are muscle length and/or the portion of the movement where the peak torque occurs. Therefore, we present the exercise selection literature in the following section.

Initial ROM vs. inter ROM and final ROM vs. inter ROM

Unfortunately, studies comparing hypertrophic responses between the initial and final ROM vs. inter ROM were not found. This represents a research gap that warrants attention from researchers.

Exercise comparisons

Ten studies comparing muscle hypertrophy after training with different resistance exercises were identified. The summarized information from these studies is presented in ▶ **Table 3**.

Kinoshita et al. [65] compared calf raises with straight vs. bent knee and observed greater hypertrophy of the two gastrocnemius heads after straight leg calf raises but similar soleus growth. These outcomes partially corroborate the findings of Burke et al. [66], who also observed greater growth in the gastrocnemius heads following straight-knee compared to bent-knee exercises, while the results were equivocal for the soleus (estimated group difference closer to zero but still favoring the bent-leg calf raise was obtained

▶ **Table 3** Summary of studies comparing the hypertrophic effects of different resistance exercises.

Study	Sample	Training protocol	Exercise	Muscles assessed	Overall hypertrophy effect
Burke et al. [66]	Trained young adults	8 wks, 2x/wk, 4 sets of 8–12RM	Leg press and straight-leg toe press × leg press and seated calf raise × leg extension and straight-leg toe press × leg extension and seated calf raise	Muscle thickness of Rectus femoris and vastus lateralis at 30 %, 50 %, and 70 % of femur length; and the medial gastrocnemius, lateral gastrocnemius and soleus muscles.	Leg press training led to greater increases in the vastus lateralis at all sites compared to leg extension training. In contrast, leg extension training resulted in more favorable increases in the rectus femoris at all sites compared to leg press. Similarly, straight-leg toe raises promoted greater hypertrophy in both the lateral and medial gastrocnemius compared to seated calf raises, while seated calf raises induced more favorable growth in the soleus muscle.
Kinoshita et al. [65]	Untrained young adults	12 wks, 2x/wk, 3–5 sets, 10 reps at 70 % 1RM	Seated calf raise vs. standing calf raise	Volume of whole and individual heads of triceps surae	Standing calf raise induced greater increases in lateral (standing: 12.4 %, seated: 1.7 %) and medial (standing: 9.2 %, seated: 0.6 %) gastrocnemius head, but similar changes in the soleus (standing: 2.1 %, seated: 2.9 %).
Larsen et al. [69]	Untrained men	10 wks, 2x/wk, 4 sets of 15–20RM	Leg extension with hip flexed at 40° × 90°	Muscle thickness of rectus femoris and vastus lateralis at 50 % and 70 % of femur length	Knee extension at 40° compared to 90° of hip flexion induced greater rectus femoris at 50 % and 70 % of length (40°: 12.4 % and 15.8 %; 90°: 4.6 % and 10.9 %). For the vastus lateralis, both training groups presented similar growth at 50 % and 70 % length (40°: 10.8 %, and 9.9 %; 90°: 9.7 %, and 11.8 %).
Larsen et al. [73]	Resistance-trained young and middle-aged men and women	8 wks, 2x/wk, 5 sets of 12–16RM	Cable lateral raise vs. dumbbell lateral raise	Muscle thickness of lateral deltoid at proximal and distal sites	Similar increases were seen between cable × dumbbell lateral raise at distal (4.6 % and 3.9 %, respectively) and proximal sites (3.3 % and 3.4 %)
Maeo et al. [70]	Untrained young adults	12 wks, 2x/wk, 5 sets of 10 reps at 50 %–70 % 1RM	Elbow extension with neutral arm vs. overhead arm	Volume of whole and individual heads of triceps brachii	Overhead arm induced greater increases of whole (overhead arm: 19.9 %, neutral arm: 13.9 %), long (overhead arm: 28.9 %, neutral arm: 19.6 %), and lateral + medial (overhead arm: 14.6 %, neutral arm: 10.5 %)
Maeo et al. [32]	Untrained young adults	12 wks, 2x/wk, 5 sets of 10 reps at 50 %–70 % 1RM	Prone leg curl vs. seated leg curl	Volume of whole and individual heads of hamstrings CSA of biceps femoris long head and semitendinosus, both measured at proximal and distal site	Seated-leg curl induced greater volume increase of whole (seated: 14.1 %, prone: 9.3 %), biceps femoris long head (seated: 14.4 %, prone: 6.5 %), semitendinosus (seated: 23.6 %, prone: 19.3 %), and semimembranosus (seated: 8.2 %, prone: 3.6 %); conversely, greater changes in sartorius were observed after prone leg curl (seated: 7.8 %, prone: 11.8 %) Seated leg curl induced a greater increase of CSA of biceps femoris long head at proximal (seated: 20.8 %, prone: 8.7 %), and distal (seated: 10.7 %, prone: 5.4 %) sites, and semitendinosus proximal (seated: 28.2 %, prone: 21.1 %), but not at the distal site (seated: 21.4 %, prone: 17.0 %)

▶ **Table 3** Continued.

Study	Sample	Training protocol	Exercise	Muscles assessed	Overall hypertrophy effect
Maeo et al. [67]	Untrained young men	12 wks of eccentric training, 2–3x/wk, 2–4 sets of 6–10 reps to failure	Nordic vs. seated leg curl	Volume of whole and individual heads of hamstrings (knee flexors) and also of sartorius, gracilis and popliteus (hip extensors)	The seated leg curl induced greater muscle growth compared to the Nordic exercise in the biceps femoris long head (seated: ~18%, Nordic: ~5%) and in the semimembranosus (seated: ~14%, Nordic: ~2%). Conversely, the Nordic exercise resulted in greater hypertrophy in the biceps femoris short head (seated: ~6%, Nordic: ~22%) and in the sartorius (seated: ~8%, Nordic: ~18%). Both exercises produced similar growth in the semitendinosus (seated: ~27%, Nordic: ~20%) and gracilis (seated: ~24%, Nordic: ~29%). When the muscle volume was combined, the seated leg curl resulted in greater growth in knee flexors and hip extensors (seated: ~19%, Nordic: ~9%), while the Nordic exercise led to greater growth in hip extensors alone (seated: ~11%, Nordic: ~21%). Overall, the seated leg curl induced greater hypertrophy in the posterior thigh muscles compared to the Nordic exercise (17% vs. 13%).
Nunes et al. [72]	Untrained young adults	10 wks, 3x/wk, 3 sets of 8–12RM	Cable preacher curl vs. barbell preacher curl	Muscle thickness of elbow flexors at the middle site	Similar increases were observed after cable and barbell preacher curl (cable: 7.0%, barbell: 8.0%)
Stasinaki et al. [71]	Untrained young women	6 wks, 2x/wk, 6 sets of 6RM	Push-down vs. overhead elbow extension	Muscle thickness of the long head of triceps brachii at proximal and distal sites CSA of the long-head triceps proximal, distal, and total	Similar increases in muscle thickness at proximal (push-down: 10.7%, overhead: 13.7%) and distal (push-down: 15.5%, overhead: 19.4%) sites Similar increases in the total (push-down: 15.3%, overhead: 16.1%) and distal (push-down: 16.7%, overhead: 24.9%) sites, but no pre to post changes at the proximal site in either group (push-down: 13.5%, overhead: −0.1%)
Zabaleta-Korta et al. [8]	Resistance-trained young women	9 wks, 3x/wk, 4 sets of 12RM	Inclined biceps curl vs. preacher biceps curl	Muscle thickness of elbow flexors at proximal, middle, and distal sites	Only the preacher curl group increased the distal site of the elbow flexors (inclined: 2.6%, preacher: 9.7%), without changes in the proximal (inclined: 2.3%, preacher: 6.6%) and middle (inclined: 4.8%, preacher: 7.8%) sites.

Abbreviations: wks = weeks, RM = repetition maximum, CSA = cross-sectional area.

for the soleus) [66]. Such findings may be interpreted as proof of the concept that biarticular muscles (i. e., gastrocnemii), when shortened at both origin and insertion, may not be able to produce relevant force and consequently experience less muscle growth. On the other hand, resistance exercises that require the muscle to produce force at longer lengths seem to optimize muscle growth. In line with this reasoning, Maeo et al. [67] compared the hypertrophy of the hip extensors and knee flexors after 12 weeks of training using the Nordic exercise versus seated leg curl. The main results showed that the seated leg curl led to greater hypertrophy in the combined muscles, while the Nordic exercise resulted in more pronounced hypertrophy when the analysis was restricted to the hip extensors. Note that these results may have been influenced by the threefold hypertrophy of the biceps femoris long head and semitendinosus from the seated leg curl compared to the Nordic exercise. Kellis and Blazevich [68] suggest that when the hamstrings are in a lengthened position, these two muscles contribute significantly more to knee flexion torque production than the other two. The results of Maeo et al. [67] with the assumption of Kellis and Blazevich [68] suggest the importance of internal muscle force production of individual muscles in determining which muscle within a muscle group will experience greater hypertrophy; this assumption follows the principle of neuromechanical matching (70).

In another study, Maeo et al. [32] compared the effects of the seated leg curl vs. prone leg curl on whole and individual hamstring muscles and observed a greater increase in biarticular muscles (i. e., biceps femoris long head, semimembranosus, semitendinosus) after the seated leg curl. Interestingly, the prone leg curl induced

a greater sartorius increase and this muscle is probably at a longer (or optimal) length when the hip is extended rather than flexed. If we assume that the external torque provided by both seated and prone leg curl machines was the same, such findings also support the notion that training the muscles in a lengthened position appears to be related to greater muscle growth.

In line with this, Larsen et al. [69] compared the hypertrophy of the rectus femoris and vastus lateralis after a training period using knee extension exercises at 90º vs. 40º of hip flexion (0 ° = full hip extension, thigh aligned with the trunk). As expected, similar growth was observed in the vastus lateralis, as its muscle length is not affected by hip joint position due to its monoarticular nature. However, rectus femoris hypertrophy was greater with 40º of hip flexion compared to 90º. The reduction in hip flexion angle increases the length of the rectus femoris, leading perhaps to greater mechanical tension experienced by the muscle. This, in theory, helps to explain the enhanced hypertrophic response of the rectus femoris when trained at longer muscle lengths.

Maeo et al. [70] compared performing elbow extension with the arm in the overhead vs. in the neutral position (anatomical position) and observed greater hypertrophy of the long and lateral + medial triceps brachii heads after overhead elbow extension. Once again, the greater growth of the long triceps brachii head suggests that training at longer muscle lengths plays a role in the magnitude of muscle growth adaptation. However, following this rationale, the greater increase in the lateral + medial heads is unexpected. Given that these muscles cross only the elbow joint, it is possible to speculate that the muscle lengths of the lateral and medial heads do not differ substantially between the exercises. Therefore, factors other than muscle length may have modulated the muscle growth of triceps brachii monoarticular heads. In apparent contrast to these findings, Stasinaki et al. [71] performed the same comparison—i. e., elbow extension with arms overhead vs. the neutral position—and observed no differences in hypertrophy of the triceps brachii long head growth. Such findings are difficult to reconcile, as such dissimilarities may be related to aspects such as muscle measurement (volume vs. muscle thickness and cross-sectional area), duration of intervention (12 weeks vs. 6 weeks), or the exercise form (e. g., distance of the participant to the pulley, height of the cable, factors that may affect the portion of the ROM where peak torque occurs).

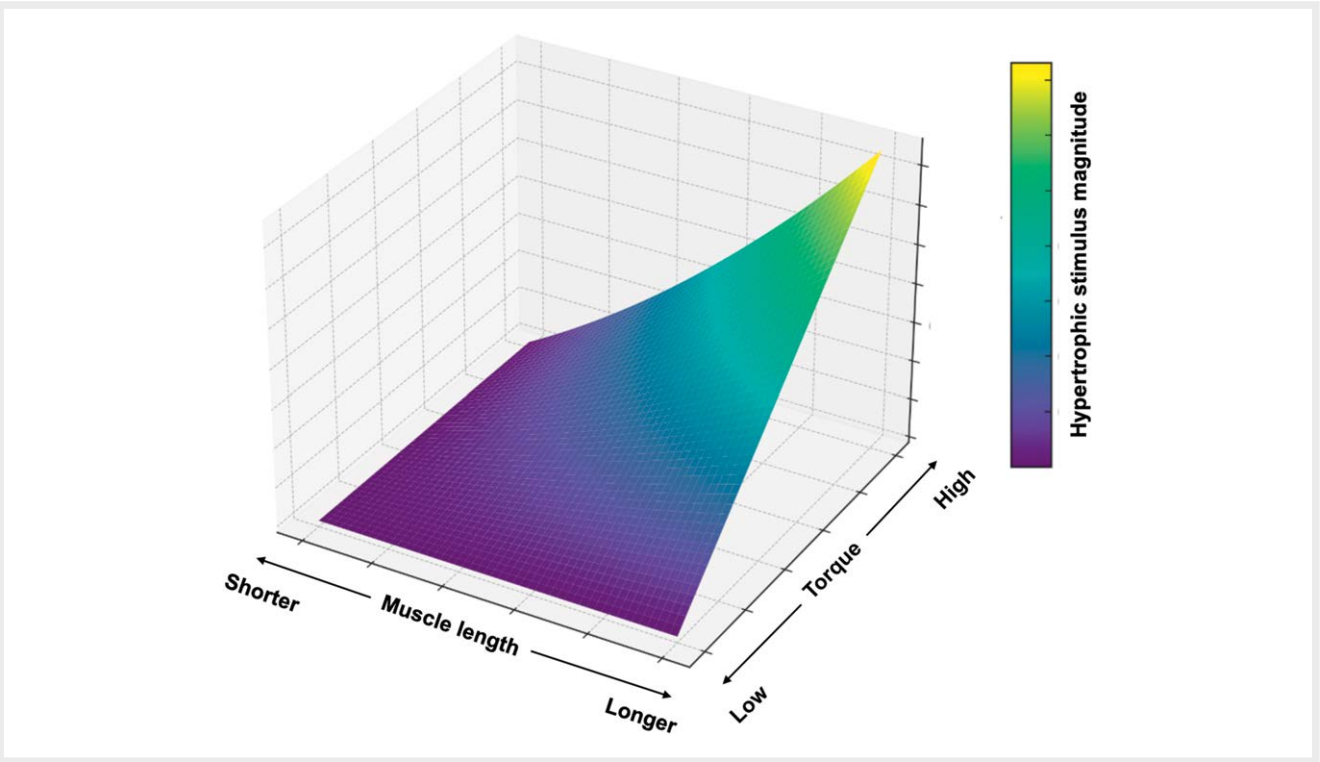
Similar to what we argued in the ROM section, it appears that longer muscle length is not the only factor that could explain the greater hypertrophy but actually the interaction between muscle length and external torque. Zabaleta-Korta et al. [8] compared the effects of the inclined biceps curl (longer muscle length) vs. the preacher biceps curl (shorter muscle length) in elbow flexor muscle size. The authors did not observe differences in the proximal and middle sites; actually, the preacher biceps curl exercise induced a more favorable increase in the distal site. When analyzing the external torque, it is possible to notice that in the inclined biceps curl there was probably no relevant external torque in the most lengthened position (initial portion of ROM) [8]. Based on this, long muscle length per se was not sufficient to induce more favorable hypertrophy; it appears that the combination of training at longer lengths with relevant external torque appears necessary to optimize muscle growth.

This notion is reinforced by the study by Nunes et al. [72], who compared preacher curls that differed in the type of external resistance (cable vs. barbell) and found no difference in elbow flexor hypertrophy between the two resistance exercises. Cable resistance likely imposed peak torque in the intermediate portion of the ROM while barbell resistance imposed peak torque at the initial portion of the ROM [72]. Notably, given the shoulder flexion in the preacher curl, it can be considered shorter muscle-length training. Thus, peak torque at shorter muscle lengths was not sufficient to induce superior growth. However, additional studies are still needed to better address this hypothesis because there is still no consensus, as reflected in the results of another study by Larsen et al. [73]. This study showed that exercise with peak torque at a greater muscle length (cable lateral raise) promoted similar hypertrophy in the distal and proximal regions of the lateral deltoid compared to exercise with peak torque at a shorter muscle length (dumbbell lateral raise) [73]. From these results, ROM and exercise selection literature indicate that the combination of relevant external torque at longer muscle lengths appears to be the driver to induce greater muscle hypertrophy and that such isolated factors may not be sufficient.

Gaps, limitations, and research recommendations

Our review pieces together different puzzles (isometric training at different joint angles, ROM, and exercise selection literature) to provide insights and draw practical conclusions. Despite this, the present study has limitations and the literature has gaps that deserve attention. First, some would argue that resistance exercises differ by more factors than muscle length and external torque. Indeed, the exercise selection literature has several nuances; for example, resistance exercises may differ in stability requirement, complexity, and number of muscles involved. All of these factors to some degree may affect the hypertrophic response. Of note, by limiting our analysis to head-to-head comparisons between single-joint exercises, the influence of these factors is attenuated. In this context, we believe that the exercise selection literature serves as a proof of concept of the interaction between muscle length and external torque, and together with isometric training and ROM literature, allows discussions on the topic to advance. Second, it has been suggested that structural adaptations (e. g., increased fascicle length, sarcomerogenesis) occur in the muscle after training at longer muscle lengths [74], theoretically attenuating the contribution of passive structures to force production in future training sessions. If accepted, this hypothesis implies that the additional benefits of training the muscle at lengthened positions may diminish after a period of training. In fact, most of the studies included in this review were conducted with untrained individuals (except for three [8, 29, 60]) and for a short period that varied from 5 to 15 weeks. Therefore, studies are needed to determine whether the strategy of training at a longer length continues to be more effective after a longer period of training and also whether resistance-trained individuals benefit from this strategy.

Third, some muscles have not yet been investigated, such as the pectoral, latissimus dorsi, and trapezius. Thus, investigations of such muscles as well as replication studies of the muscles already investigated will be great additions to the specialized literature. Fourth, longer muscle-length contractions often induce a greater



▶ **Fig. 1** Conceptual model illustrating the qualitative interaction between muscle length and external torque in determining hypertrophic stimulus. It represents a hypothetical hypertrophic stimulus resulting from the interaction of muscle length (x-axis) and torque (y-axis). The z-axis reflects the relative magnitude of hypertrophic stimulus. The model suggests that while training at longer muscle lengths generally favors muscle growth, it is not sufficient on its own. The combination of moderate to longer muscle lengths with meaningful external torque applied in that muscle position appears to be critical for maximizing hypertrophic adaptations.

reduction in performance (e. g., peak force) and reduced muscle excitation immediately after a training session and in the recovery period (up to 48 h) [52]. This raises the question of whether training at longer muscle length can negatively impact a subsequent session targeting the same muscle group; particularly when training the same muscle group with high frequency and/or or high set volume. Such factors require investigation. Fifth, the fitness industry provides types of equipment that allow the manipulation of the portion of the ROM where the torque peak occurs, with machines that make it possible by changing the position of the weight plates or by changing the cam radius. Of note, such apparatus remains less explored by scientific literature, therefore its effectiveness in modifying hypertrophic responses is unknown and requires investigation. Finally, we focus our argument on the interaction between muscle length and external torque. High external torque does not necessarily correspond to high internal muscle force. For example, increasing the velocity at the beginning of the concentric phase seems to shift the peak internal torque from the end to the beginning of the movement during the knee extension exercise [75]. It has been suggested that the internal moment arm length may be a factor affecting which muscle will contribute most to producing torque at the joint, following the principle of neuromechanical matching [76, 77]. In theory, it affects which muscle experiences the greatest mechanical tension and muscle hypertrophy in response to training at a specific muscle length, ROM, or exercise selection. Notably, this hypothesis needs empirical support.

Conclusions

From different lines of investigation, it is possible to infer that muscle length plays a role in the magnitude of muscle hypertrophy; achieving moderate to longer muscle lengths appears to be important for inducing appreciable muscle growth. Of note, muscle length appears not to be the only factor. From the results, there seems to be an interaction between muscle length and external torque[1]. The superior muscle growth often observed after exercises and ROMs that train muscles at longer lengths occurs particularly when there is significant external torque in the lengthened position. This inference is made from the following observations: 1) full and initial ROM generally combine training at longer muscle lengths with high external torque in that position and induce more muscle growth; 2) despite high external torque, training at shorter muscle lengths, such as final ROM, fails to optimize muscle growth; and 3) exercises that train the muscle in a lengthened position with reduced torque in that position fail to induce greater gains. Collectively, these lines of investigation suggest that training at longer muscle lengths is necessary but not sufficient to maximize muscle hypertrophy. The combination of training at moderate to long muscle lengths with relevant external torque in that position seems to

[1] We would like to highlight that external torque is affected by the load lifted and the external moment arm. The arguments presented by us about external torque apply in contexts where the intensity used (e. g., % of 1RM or repetition range) is the same between experimental conditions.

be driving this adaptation. We have provided a figure to aid in the visualization of the hypothetical relationship between muscle length and external torque (▶**Fig. 1**).

Practical considerations

Greater physiological responses linked to muscle hypertrophy are often observed after contractions performed at moderate and longer muscle lengths. These acute responses agree with the differential adaptations in response to muscle length, ROM, and exercise selection found in this review. From a practical perspective, isometric training at moderate to longer muscle lengths induces more favorable hypertrophy, therefore they should be considered at the expense of isometric training at shorter muscle lengths. The different ROM configurations seem not to be the main factor, but a means of manipulating the length that the muscle is trained in and the magnitude of external torque that will be imposed. Therefore, strength and conditioning coaches and practitioners do not need to adhere to a specific ROM when the objective is to optimize muscular hypertrophy; rather our suggestion is to see the different ROM configurations as a strategy to train the muscles at longer lengths with relevant external torque. Accordingly, the selection of exercises for hypertrophy-oriented programs should consider the inclusion of resistance exercises that impose relevant torque at longer muscle lengths.

Conflict of Interest

The authors declare that they have no conflict of interest.

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